

- ▶ KVM stands for Kernel-based Virtual Machine. It's a Linux kernel module.
- ▶ Qemu, although it can work on Windows, is more suitable for Linux.

We call Qemu/KVM as Linux-only virtualization because Qemu in itself performs poorly without KVM and with better options available, perhaps no one would want to use Qemu. KVM is the Linux kernel module which helps a virtual machine leverage the benefits of the virtualization assistive technologies built into modern processors (Intel VT-X and AMD-D). Combining KVM and Qemu, one gets a nice set of features with great performance. It's this increase in performance which led to the development of a separate branch of Qemu called Qemu-KVM. It combined and integrated Qemu with KVM and performs a lot better. As such, this section is only for Linux guys.

Installation

Installing Qemu and KVM is not very difficult. Just use your package manager to install the packages `kvm`, `qemu-kvm` and `virt-manager`. There are a lot of those packages which will install themselves as dependencies. The total download size should be between 10 and 30 MB, depending on the packages already installed.

Usage

Using Qemu/KVM is not as easy as the other two virtualization software we've discussed. You need the command line. If luck favoured you and `virt-manager` package was installed properly, you might just run that and get your good share of ease-of-use. The `virt-manager` package is actually GUI-based and uses a library (named "libvirt") to command Qemu. If you have it installed and working correctly, use it to create a new virtual machine. The wizard won't be difficult to follow for it lacks most features compared to other virtualization software we talked about till now.

There are quite a few chances that `virt-manager` won't behave as you might expect it to. We found errors that varied from one Linux distribution to another. While in Fedora, it complained about not finding KVM when we tried to create a new virtual machine. It asked us to use only the root account on openSUSE and simply dumped a segmentation fault on Ubuntu! So here's how to go about Qemu with the command line.

Before anything, you must load the KVM module into the kernel. For this, issue the command as root:

```
modprobe kvm
```

